

Problem Set #4: Classical Mechanics (PH31207)

August 31, 2026

Lagrangian and Euler-Lagrange Equations

1. Write down the Lagrangian for an Atwood's machine. Obtain the accelerations of the two masses using the Euler-Lagrange equation.
2. Consider a simple pendulum length l attached to a moving lift falling with an acceleration f . Using the Lagrangian formalism, show that an observer in the moving lift measures the time period of the pendulum as $2\pi\sqrt{\frac{l}{g-f}}$. Here, of course, we assume all the velocities to be small enough so that one can neglect relativistic effects.
3. Write the Lagrangian for the central field using the spherical polar coordinate. Then calculate all generalized momenta. What can you say about p_ϕ ?
4. In this problem, you derive the expression for the generalized potential for the electromagnetic forces of moving charges.
 - (a) Using Gauss' law (for magnetism) $\vec{\nabla} \cdot \vec{B} = 0$ and Lenz's law $\vec{\nabla} \times \vec{E} + \frac{\partial \vec{B}}{\partial t} = 0$, write down the electric and magnetic fields in terms of a scalar potential ϕ and a vector potential $\vec{A}$.
 - (b) Writing the Lorentz force law in terms of potentials ϕ and $\vec{A}$, prove that it is possible to write $\vec{F} = -\vec{\nabla}U + \frac{d}{dt}(\vec{\nabla}_{\vec{v}}U)$ for $U = q\phi - q(\vec{v} \cdot \vec{A})$.
 - (c) Write a suitable Lagrangian for the charged particle moving under an electromagnetic field. What will be the particle's generalized momenta?
5. Consider a particle moving with a velocity $\vec{v}$ in a rotating frame S' . The frame S' rotates with an angular velocity $\vec{\omega}$ with respect to an inertial frame S . There is a scalar potential $V(\vec{r})$ in S , where $\vec{r}$ denotes a particle's position vector expressed in the rotating basis.
 - (a) Write down the Lagrangian for the system. Write down the expression of the effective potential as seen by an observer in S' .
 - (b) In the Euler-Lagrange equations, what does the term that is quadratic in ω signify? What is its direction – centripetal or centrifugal? Is it a pseudo or a real force?